

NAME: _____

PERIOD: _____

DATE: _____

One Plot, Two Reporting Strategies

AP Statistics · Unit 5 · Topic 5.1 · B: 2027-03-02 · E: 2027-03-05

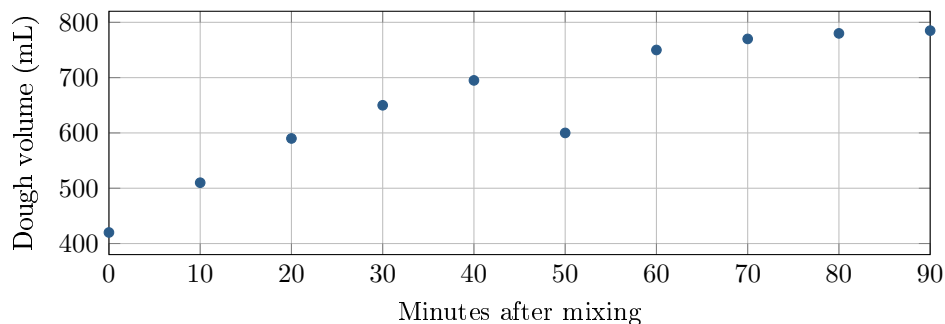
[STAR] TODAY'S PROBLEM

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

Jules: Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

Ravi: Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.

Selected measurements (minutes, mL): (0, 420), (20, 590), (70, 770), (90, 785).



FIRST TAKE — before we discuss (5 min)

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.

[BOOK] BUILD A COMPLETE SCATTERPLOT DESCRIPTION (keep on the page)

Put the explanatory variable on the x -axis and the response on the y -axis. Describe direction, form, strength, and unusual features in context. Linear form has an approximately constant rate; nonlinear form may curve or change rate. Report an unusual point without letting that one point erase the pattern followed by the rest.

Work (a)–(c) from the rules box:

DOK 1 (a) Name the explanatory and response variables, with units.

DOK 2 (b) Find the volume increase from 0 to 20 minutes and from 70 to 90 minutes. Is the 50-minute batch above or below the nearby batches?

DOK 3 (c) Whose reporting strategy is supported? Describe the direction, form, strength, and unusual point in context, using your two increases. Explain how a straight-line summary could mislead someone predicting later growth. Write 3–4 sentences.

Frame: I choose _____: the pattern is _____ in direction, _____ in form, and _____ in strength.

Frame: The two increases, _____ and _____ mL, show _____; a straight line could mislead by _____.

EXIT — turn this in

One thing the rules box changed about my first take: _____